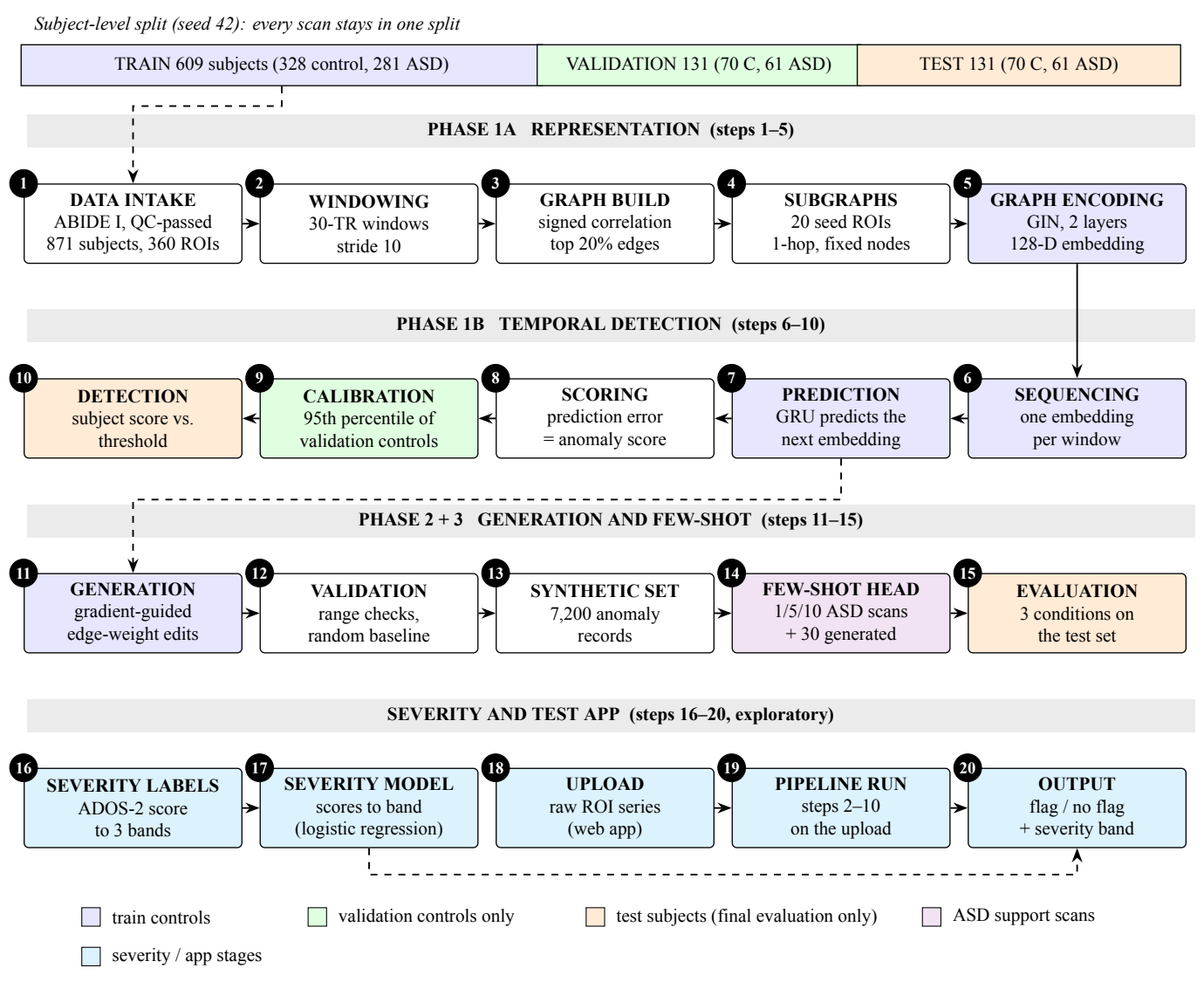

Subject-level split (seed 42): every scan stays in one split
TRAIN 609 subjects (328 control, 281 ASD)
VALIDATION 131 (70 C, 61 ASD)
TEST 131 (70 C, 61 ASD)
PHASE 1A REPRESENTATION (steps 1–5)
1 DATA INTAKE
ABIDE I, QC-passed
871 subjects, 360 ROIs
2 WINDOWING
30-TR windows
stride 10
3 GRAPH BUILD
signed correlation
top 20% edges
4 SUBGRAPHS
20 seed ROIs
1-hop, fixed nodes
5 GRAPH ENCODING
GIN, 2 layers
128-D embedding
PHASE 1B TEMPORAL DETECTION (steps 6–10)
6 SEQUENCING
one embedding
per window
7 PREDICTION
GRU predicts the
next embedding
8 SCORING
prediction error
= anomaly score
9 CALIBRATION
95th percentile of
validation controls
10 DETECTION
subject score vs.
threshold
PHASE 2 + 3 GENERATION AND FEW-SHOT (steps 11–15)
11 GENERATION
gradient-guided
edge-weight edits
12 VALIDATION
range checks,
random baseline
13 SYNTHETIC SET
7,200 anomaly
records
14 FEW-SHOT HEAD
1/5/10 ASD scans
+ 30 generated
15 EVALUATION
3 conditions on
the test set
SEVERITY AND TEST APP (steps 16–20, exploratory)
16 SEVERITY LABELS
ADOS-2 score
to 3 bands
17 SEVERITY MODEL
scores to band
(logistic regression)
18 UPLOAD
raw ROI series
(web app)
19 PIPELINE RUN
steps 2–10
on the upload
20 OUTPUT
flag / no flag
+ severity band
train controls
validation controls only
test subjects (final evaluation only)
ASD support scans
severity / app stages